

Estimated Hours

Calculate Use Case Weight (UUCW)

Use case: "Configure Game" (Console input, mine count selection, validation)

Complexity: Simple

Weight: 5 points

Use case: "Uncover Tile" (Click, dynamic mine placement, recursive dig, win/loss check)

Complexity: Average

Weight: 10 points

Use case: "Toggle Flag" (Right click and update HUD counter)

Complexity: Simple

Weight: 5 points

Use case: "Restart Game" (End screen click, board reset, variable reinitialization)

Complexity: Average

Weight: 10 points

Total UUCW = 5 + 10 + 5 + 10 = 30

Calculate Actor Weight (UAW)

Actor: "Player" (Interacts via graphical mouse interface)

Complexity: Complex

Weight: 3 points

Actor: "System" (Pygame event loop and window management)

Complexity: Simple

Weight: 1 point

Total UAW = 3 + 1 = 4

Calculate Unadjusted UCP (UUCP)

= UUCW + UAW = 30 + 4 = 34

Final Effort

Assume a 20% adjustment factor (1.20 multiplier) for recursive flood-fill and Pygame graphics handling, plus full board-state reset logic:

Adjusted UCP = $34 \times 1.20 = 40.8$

Assume a productivity rate of 3 person-hours per UCP due to the team's size and small scale of the project:

Final Estimated Effort = $40.8 \times 3 = 122.4$ person-hours